Code No: 133BM **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, November/December - 2018
OBJECT ORIENTED PROGRAMMING THROUGH JAVA
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1a)	What is inheritance? Give example.	[2]
b)	Define the basic characteristics of object oriented programming.	[3]
c)	What is Console class? What is its use in java?	[2]
d)	What is the use of auto boxing in java? Explain.	[3]
e)	What is thread based preemptive multitasking?	[2]
f)	How do we set priorities for threads?	[3]
g)	What is a Java Priority queue?	[2]
h)	What is a Collection Class? Give an example.	[3]
i)	What is Swing in Java? How it differs from Applet.	[2]
j)	How do applets differ from application program?	[3]

PART-B

(50 Marks)

2.a) Describe the structure of a typical Java program with an example.
b) Write the significance of Java Virtual Machine. [5+5]

OR

3.a) How do we implement polymorphism in JAVA? Explain briefly.
b) What is an array? How do you declare the array in java? Give examples. [5+5]

4.a) How to design and implement an interface in Java? Give an example.
b) Give an example where interface can be used to support multiple inheritance. [5+5]

OR

5.a) What are the methods available in the Character Streams? Discuss.
b) Distinguish between Byte Stream Classes and Character Stream Classes. [5+5]

6.a) What is an Exception? How is an Exception handled in JAVA?
b) Write a java program that illustrates the application of multiple catch statements. [5+5]

OR

7.a) Differentiate between multiprocessing and multithreading. What is to be done to implement these in a program?
b) Write a program that creates two threads. Fist thread prints the numbers from 1 to 100 and the other thread prints the numbers from 100 to 1. [5+5]

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8. a) What are similarities and difference between ArrayList and Vector? Explain.
b) What is different between Iterator and ListIterator? Explain different ways to iterate over a list. [5+5]

OR

9.a) What are the best practices related to Java Collections Framework? Discuss.
b) What is Comparable and Comparator interface? Differentiate between them. [5+5]

10.a) What is the difference between init( ) and start ( ) methods in an Applet? When will each be executed?
b) Write the applets to draw the Cube and Circle shapes. [5+5]

OR

11.a) Explain various layout managers in JAVA.
b) Write a program to create a frame window that responds to mouse clicks. [5+5]

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